

Question

How many stereoisomers does monofluoromonochloromonobromomonoiodocubane have? Optical isomers must be considered.

A. 6

B. 7

C. 10

D. 20

E. 28

F. 32

G. 38

H. 48

I. 57

J. 64

K. 70

L. 75

M. 82

N. 88

O. 93

P. 100

Answer

Correct Answer: **K**

Detailed Explanation

First, consider the number of isomers when the four halogen atoms are identical:

- (1) All 4X on the same face: 1,2,3,4- — 1 type
- (2) Three X on one face: cases include 1,2,3,5- / 1,2,3,6- / 1,2,4,5- — 3 types
- (3) Two X on the same face: cases include 1,2,5,6- / 1,3,5,7- — 2 types

Thus, there are a total of 6 possible arrangements.

CHECKPOINT

1 PTS

Four identical halogen atoms have 6 possible arrangements

Next, calculate the number of isomers for monofluoro-monochloro-monobromo-monoiodo cubane under these cases:

- (1) 1,2,3,4-: Fix the F atom, and there are 3 possibilities for the diagonal positions, each with optical isomers — total: $3 \times 2 = 6$ types

CHECKPOINT

1 PTS

1,2,3,4- has 6 types

(2) 1,2,3,5-: Position 5 has 4 possibilities, position 1 has 3, all with optical isomers — total: $4 \times 3 \times 2 = 24$ types

CHECKPOINT

1 PTS

1,2,3,5- has 24 types

(3) 1,2,3,6-: Position 1 has 4 possibilities, all with optical isomers — total: $4 \times 2 = 8$ types

CHECKPOINT

1 PTS

1,2,3,6- has 8 types

(4) 1,2,4,5-: The two adjacent atoms have C_4^2 possibilities, and the remaining two have 2 possibilities, all with optical isomers — total: $6 \times 2 \times 2 = 24$ types

CHECKPOINT

1 PTS

1,2,4,5- has 24 types

(5) 1,2,5,6-: Fix the F atom, adjacent positions have 3 possibilities, face-diagonal positions have 2, with no optical isomers — total: $3 \times 2 = 6$ types

CHECKPOINT

1 PTS

1,2,5,6- has 6 types

(6) 1,3,5,7-: 2 optical isomers — 2 types

CHECKPOINT

1 PTS

1,3,5,7- has 2 types

Thus, there are a total of 70 types.

CHECKPOINT

1 PTS

There are 70 types of isomers in total